PanicSell -- Technical Architecture Document

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1. Project Overview

PanicSell is a single-user, multi-broker stock liquidation tool built for the Indian equity market. It provides a unified dashboard to view holdings from Zerodha and Groww, and execute sell orders instantly -- individually, per-broker, or across all connected brokers simultaneously.

Purpose

During market panic events, manually selling stocks across multiple broker apps is slow and error-prone. PanicSell consolidates this into a single interface with one-click sell capabilities, AMO (After Market Order) support, and automated portfolio drop alerts.

Scope

- Personal-use tool (single user, single backend instance)
- Read-only data access (holdings, margins, orders, quotes) and sell-order execution
- No buy orders, no algorithmic trading, no multi-user tenancy
- Compliant with SEBI 2025 regulations for personal automated trading

Tech Stack

Layer	Technology
Frontend	React 18, TypeScript, Vite 6, Tailwind CSS 3
Backend	Python 3.12, FastAPI, Uvicorn
Broker SDKs	`kiteconnect` (Zerodha), `growwapi` (Groww), `pyotp` (TOTP)

Layer	Technology
Email	Gmail SMTP via `smtplib` (SSL, port 465)
Container	Docker (`python:3.12-slim`)
Frontend Hosting	Vercel (static CDN + API rewrites)
Backend Hosting	Render (persistent Docker container)

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2. System Architecture

High-Level Topology

```
graph LR
    Browser["User Browser"] -->|"HTTPS"| Vercel["Vercel CDN<br/>(React SPA)"]
    Vercel -->|"/api/* rewrite"| Render["Render<br/>(FastAPI + Docker)"]
    Render -->|"OAuth redirect + API"| Zerodha["Zerodha<br/>Kite Connect API"]
    Render -->|"TOTP + API"| Groww["Groww<br/>Trading API"]
    Render -->|"SMTP SSL"| Gmail["Gmail<br/>SMTP Server"]
    Zerodha -->|"OAuth callback"| Render
```

Separation of Concerns

The system is split into two independently deployed services:

- **Vercel (Frontend)**: Serves the static React SPA from a global CDN. All `/api/*` requests are rewritten at the edge to the Render backend. No server-side logic, no environment variables, no state.
- **Render (Backend)**: Runs a persistent Docker container with the FastAPI application. Holds in-memory broker sessions, runs background alert monitoring, and communicates with broker APIs and Gmail SMTP.

The frontend never communicates directly with broker APIs. All broker interactions are proxied through the backend, which holds the access tokens.

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3. Deployment Infrastructure

Frontend -- Vercel

Property	Value
Platform	Vercel
Framework	Vite (auto-detected)
Build Command	`npm run build`
Output Directory	`dist/`
Root Directory	`frontend/`
URL	`https://panicsell-nine.vercel.app`

API Proxy -- configured in `frontend/vercel.json`:

```
{
  "rewrites": [
    {
      "source": "/api/:path*",
      "destination": "https://panicsell-backend.onrender.com/api/:path*"
    }
  ]
}
```

All requests matching `/api/*` are rewritten to the Render backend at the edge. The browser sees same-origin requests, avoiding CORS issues on the client side.

Backend -- Render

Property	Value
Platform	Render
Service Type	Web Service (Docker)
Instance	Free tier (sleeps after 15 min inactivity)
Root Directory	`backend/`
URL	`https://panicsell-backend.onrender.com`
Port	Injected via `$PORT` env var (default: 10000 on Render)

Dockerfile (`backend/Dockerfile`):

```
FROM python:3.12-slim
WORKDIR /app
RUN apt-get update && apt-get install -y --no-install-recommends gcc && rm -rf /var/lib/apt/lists/*
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY . .
CMD uvicorn app.main:app --host 0.0.0.0 --port ${PORT:-8000}
```

The `gcc` package is required for compiling C extensions in `kiteconnect` and `growwapi`.

Build exclusions (`.dockerignore`): `venv/`, `__pycache__/`, `*.pyc`, `.env`, `.git/`

Environment Variables (Render Dashboard)

Variable	Purpose	Required
`KITE_API_KEY`	Zerodha Kite Connect API key	Yes (for Zerodha)
`KITE_API_SECRET`	Zerodha Kite Connect API secret	Yes (for Zerodha)
`GROWW_TOTP_TOKEN`	Groww TOTP token for auto-login	Recommended
`GROWW_TOTP_SECRET`	Groww TOTP secret for code generation	Recommended
`GROWW_API_KEY`	Groww API key (fallback auth)	Optional

Variable	Purpose	Required
`GROWW_API_SECRET`	Groww API secret (fallback auth)	Optional
`FRONTEND_URL`	Vercel frontend origin for CORS	Yes
`BACKEND_URL`	Render backend URL for callback construction	Yes
`SMTP_EMAIL`	Gmail address for alert emails	Optional
`SMTP_APP_PASSWORD`	Gmail app password for SMTP auth	Optional

No secrets are stored in code or committed to version control. The `.env` file is listed in both `.gitignore` and `.dockerignore`.

Auto-Deploy Pipeline

```
graph LR
    Push["git push to GitHub"] --> Vercel_Build["Vercel: npm run build"]
    Push --> Render_Build["Render: docker build"]
    Vercel_Build --> Vercel_Deploy["Vercel CDN"]
    Render_Build --> Render_Deploy["Render Container"]
```

Both Vercel and Render are connected to the same GitHub repository and trigger automatic deployments on every push to the main branch.

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4. Backend Architecture

Directory Structure

```
backend/
  app/
    __init__.py
    main.py              # FastAPI app, middleware, lifespan
    config.py            # Pydantic settings, env loading
    routers/
      __init__.py
      auth.py            # Login, callback, status, logout
      portfolio.py       # Holdings, margins, quotes
      orders.py          # Panic sell, order history
      market.py          # NSE market status
      alerts.py          # Alert config, baseline management
    services/
      __init__.py
      broker_base.py     # Abstract broker interface
      zerodha_broker.py  # Zerodha/Kite implementation
      groww_broker.py    # Groww implementation
      broker_manager.py  # Multi-broker orchestration
      email_service.py   # Gmail SMTP email sending
      alert_monitor.py   # Background alert loop
  requirements.txt
  Dockerfile
  .dockerignore
  .env.example
```

Module Dependency Graph

```
graph TD
    main["main.py<br/>(FastAPI App)"] --> auth["routers/auth.py"]
    main --> portfolio["routers/portfolio.py"]
    main --> orders["routers/orders.py"]
    main --> market["routers/market.py"]
    main --> alerts_r["routers/alerts.py"]
    main --> alert_mon["services/alert_monitor.py"]
    main --> config["config.py"]

    auth --> broker_mgr["services/broker_manager.py"]
    portfolio --> broker_mgr
    orders --> broker_mgr
    alerts_r --> broker_mgr
    alerts_r --> alert_mon
    alert_mon --> broker_mgr
    alert_mon --> email_svc["services/email_service.py"]

    broker_mgr --> zerodha["services/zerodha_broker.py"]
    broker_mgr --> groww["services/groww_broker.py"]
    broker_mgr --> base["services/broker_base.py"]
    zerodha --> base
    groww --> base
    zerodha --> config
    groww --> config
    email_svc --> config
    broker_mgr --> config
    auth --> config
```

Application Lifecycle

The FastAPI app uses an async context manager for lifespan events:

```python
@asynccontextmanager
async def lifespan(app: FastAPI):
    start_alert_loop()   # spawns background daemon thread
    yield                 # app runs
    # no explicit shutdown; daemon thread dies with process
```

CORS Configuration

```python
allowed_origins = [settings.frontend_url]
if settings.frontend_url != "http://localhost:5173":
    allowed_origins.append("http://localhost:5173")

app.add_middleware(
    CORSMiddleware,
    allow_origins=allowed_origins,
    allow_credentials=True,
    allow_methods=["*"],
    allow_headers=["*"],
)
```

In production, `allowed_origins` contains the Vercel domain. Localhost is additionally allowed to support concurrent local development. Note that Vercel rewrites bypass CORS entirely since the browser sees same-origin `/api` requests, but the CORS policy is still enforced for direct backend access.

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5. API Reference

All endpoints are prefixed with `/api`. The backend exposes 5 router groups with 14 total endpoints.

Auth (`/api/auth`)

Method	Path	Description	Auth Required
GET	`/login/{broker_name}`	Initiates broker login. For Zerodha: redirects to Kite OAuth. For Groww: generates session server-side via TOTP and redirects to frontend.	No
GET	`/callback/{broker_name}`	OAuth callback. Receives `request_token` and `status` query params, exchanges for access token, redirects to frontend.	No
GET	`/callback`	Fallback callback (assumes Zerodha). Handles legacy redirect URLs without broker name.	No
GET	`/status`	Returns connection status of all configured brokers, including profile info for connected ones.	No
POST	`/logout/{broker_name}`	Disconnects a specific broker. Invalidates Zerodha token server-side; clears Groww session in memory.	No

Portfolio (`/api/portfolio`)

Method	Path	Query Params	Description
GET	`/holdings`	`broker` (optional)	Returns holdings from one or all connected brokers, plus a computed summary (total invested, current value, P&L, stock count).
GET	`/margins`	--	Returns per-broker margins/funds and a combined total.
GET	`/quotes`	--	Returns latest LTP quotes for all holdings across all connected brokers.

Holdings Response Shape:

```
{
  "holdings": [
    {
      "tradingsymbol": "RELIANCE",
      "exchange": "NSE",
      "isin": "INE002A01018",
      "quantity": 5,
      "t1_quantity": 0,
      "total_quantity": 5,
      "average_price": 2450.00,
      "last_price": 2480.50,
      "pnl": 152.50,
      "pnl_percentage": 1.24,
      "day_change": 15.30,
      "day_change_percentage": 0.62,
      "product": "CNC",
      "is_t1": false,
      "broker": "zerodha"
    }
```

```
  ],
  "summary": {
    "total_invested": 12250.00,
    "total_current": 12402.50,
    "total_pnl": 152.50,
    "total_pnl_percentage": 1.24,
    "stock_count": 1
  }
}
```

Orders (`/api/orders`)

Method	Path	Body / Params	Description
POST	`/panic-sell`	`{ symbols?: string[], broker?: string, variety: "regular" | "amo" }`	Places sell orders. `symbols=null` sells all holdings. `broker=null` sells across all brokers. Returns per-order results with success/failure status.
GET	`/history`	`broker` (optional)	Returns today's order history, sorted by placed time (newest first).

Sell Response Shape:

```
{
  "results": [
    {
      "tradingsymbol": "RELIANCE",
      "quantity": 5,
      "order_id": "2403120001234",
      "status": "success",
      "error": null,
      "broker": "zerodha"
    }
  ],
  "summary": { "total": 1, "success": 1, "failed": 0 }
}
```

Market (`/api/market`)

Method	Path	Description
GET	`/status`	Returns NSE market status: open/closed/pre_open/post_close, AMO eligibility, current IST time, and reason.

Alerts (`/api/alerts`)

Method	Path	Body	Description
GET	`/config`	--	Returns current alert configuration (enabled, email, thresholds, baselines, cooldowns).
POST	`/config`	`{ enabled: bool, email: string, thresholds: { broker: pct } }`	Updates alert config. When enabled, snapshots current portfolio values as baselines.
POST	`/reset-baseline`	--	Re-snapshots current portfolio values as new baselines without changing other settings.

6. Authentication and Session Management

Zerodha -- OAuth 2.0 Flow

```
sequenceDiagram
    participant Browser
    participant Vercel as Vercel (Frontend)
    participant Render as Render (Backend)
    participant Kite as Zerodha Kite

    Browser->>Vercel: Click "Connect Zerodha"
    Vercel->>Render: GET /api/auth/login/zerodha
    Render->>Browser: 302 Redirect to kite.zerodha.com/connect/login
    Browser->>Kite: User enters credentials + TOTP
    Kite->>Render: GET /api/auth/callback?request_token=xxx&status=success
    Render->>Render: KiteConnect.generate_session(request_token, api_secret)
    Note over Render: Access token stored in ZerodhaBroker._access_token
    Render->>Browser: 302 Redirect to FRONTEND_URL/?connected=zerodha
    Browser->>Vercel: Load dashboard
    Vercel->>Render: GET /api/auth/status
    Render->>Vercel: { brokers: [{ broker: "zerodha", connected: true, user_name: "..." }] }
```

The Zerodha redirect URL on their developer console must point to `BACKEND_URL/api/auth/callback` (the fallback route that assumes Zerodha).

Groww -- TOTP Flow (Recommended)

```
sequenceDiagram
    participant Browser
    participant Vercel as Vercel (Frontend)
    participant Render as Render (Backend)
    participant GrowwAPI as Groww API

    Browser->>Vercel: Click "Connect Groww"
    Vercel->>Render: GET /api/auth/login/groww
    Render->>Render: pyotp.TOTP(GROWW_TOTP_SECRET).now()
    Render->>GrowwAPI: GrowwAPI.get_access_token(api_key=TOTP_TOKEN, totp=code)
    GrowwAPI->>Render: access_token
    Note over Render: Access token stored in GrowwBroker._access_token
    Render->>Browser: 302 Redirect to FRONTEND_URL/?connected=groww
```

No browser interaction with Groww is required. The TOTP code is generated server-side using the `pyotp` library, making the connection fully automatic with no daily manual approval.

Groww -- API Key+Secret Flow (Fallback)

If TOTP credentials are not configured, the backend falls back to:

```
GrowwAPI.get_access_token(api_key=GROWW_API_KEY, secret=GROWW_API_SECRET)
```

This flow works identically from the user's perspective, but requires daily manual approval on Groww's Cloud API Keys dashboard page.

Session Storage

Sessions are stored **entirely in memory** on the backend process:

Broker	Token Location	Persistence
Zerodha	`ZerodhaBroker._access_token` + `KiteConnect._access_token`	Lost on process restart
Groww	`GrowwBroker._access_token` + `GrowwAPI` instance	Lost on process restart

There is no database, no Redis, and no cookie/JWT mechanism. The `BrokerManager` is instantiated as a module-level singleton (`broker_manager = BrokerManager()`) and lives for the lifetime of the process.

Token expiry: Both Zerodha and Groww tokens expire daily (around 6-8 AM IST). Users must reconnect each morning.

Logout

- **Zerodha**: Calls `KiteConnect.invalidate_access_token()` to invalidate the token on Zerodha's servers, then clears local state.
- **Groww**: Clears in-memory state only (no server-side invalidation API available).

---

7. Broker Abstraction Layer

Abstract Interface

`backend/app/services/broker_base.py` defines the contract all broker implementations must satisfy:

Method	Signature	Return	Purpose
`is_connected`	`@property -> bool`	`bool`	Whether the broker has an active session
`get_login_url`	`() -> str`	URL or `""`	OAuth redirect URL, or empty for non-OAuth brokers
`generate_session`	`(request_token: str) -> dict`	Token dict	Exchange request token (or generate via TOTP) for access token
`get_holdings`	`() -> list[dict]`	Normalized holdings	All current holdings with prices and P&L
`place_sell_order`	`(tradingsymbol, exchange, quantity, product, variety) -> str`	Order ID	Place a market sell order
`get_margins`	`() -> dict`	Margin info	Available cash and used margin
`get_orders`	`() -> list[dict]`	Order history	Today's executed and pending orders
`get_profile`	`() -> dict`	User info	User name and ID
`get_quotes`	`(instruments: list[str]) -> dict`	LTP quotes	Latest prices for given symbols

Method	Signature	Return	Purpose
`logout`	`() -> None`	--	Clear session and invalidate tokens

Implementations

ZerodhaBroker (`zerodha_broker.py`):

- Wraps the `KiteConnect` SDK from `kiteconnect` package
- Holdings normalization: maps Kite's response to common schema, computes `total_quantity = quantity + t1_quantity`, flags `is_t1` holdings
- Sell orders: uses `VARIETY_REGULAR` or `VARIETY_AMO`, `ORDER_TYPE_MARKET`, `TRANSACTION_TYPE_SELL`
- Quotes: fetches via `kite.ltp()` for `NSE:{symbol}` instruments

GrowwBroker (`groww_broker.py`):

- Wraps the `GrowwAPI` class from `growwapi` package
- Holdings normalization: fetches LTP per-symbol via `groww.get_ltp()`, computes P&L
- Sell orders: uses `GrowwAPI.place_order()` with `TRANSACTION_TYPE_SELL`, `ORDER_TYPE_MARKET`, `PRODUCT_CNC`
- Exchange mapping: converts `"NSE"` / `"BSE"` strings to `GrowwAPI.EXCHANGE_NSE` / `GrowwAPI.EXCHANGE_BSE`
- Order status mapping: normalizes Groww statuses (`EXECUTED` -> `COMPLETE`, `ACKED` -> `OPEN`, etc.)

BrokerManager

`broker_manager.py` is the central orchestration layer:

```
graph TD
    BM["BrokerManager"] --> ZB["ZerodhaBroker"]
    BM --> GB["GrowwBroker"]
    BM --> StatusAgg["get_status() -- aggregate connection status"]
    BM --> HoldingsAgg["get_all_holdings() -- merge holdings"]
    BM --> MarginsAgg["get_all_margins() -- merge margins"]
    BM --> OrdersAgg["get_all_orders() -- merge orders"]
    BM --> SellOrch["sell_holdings() -- multi-broker sell orchestration"]
    BM --> QuotesAgg["get_all_quotes() -- merge quotes"]
```

Multi-broker sell orchestration (`sell_holdings`):

1. Determine target brokers (specific or all connected)
2. For each broker, fetch holdings
3. Filter to sellable holdings (`total_quantity > 0`)
4. If specific symbols requested, filter further
5. Place sell orders sequentially, collecting results
6. Return aggregated results with per-order success/failure

Adding a New Broker

To add a new broker (e.g., Angel One):

1. Create `backend/app/services/angelone_broker.py` implementing `BrokerBase`
2. Add credentials to `config.py` (e.g., `angelone_api_key`)
3. Register in `BrokerManager.__init__()` conditionally on credentials
4. No changes needed to routers, frontend hooks, or types -- the abstraction handles it

8. Data Architecture

Storage Model

PanicSell uses **no database**. All data falls into three categories:

Data Type	Storage	Lifetime	Location
Broker access tokens	In-memory (Python objects)	Until process restart	`ZerodhaBroker._access_token`, `GrowwBroker._access_token`
Alert configuration	JSON file on disk	Persistent across restarts	`backend/alerts_config.json`
Holdings, orders, margins, quotes	None (fetched in real-time)	Per-request	Broker APIs

Alert Configuration File

`backend/alerts_config.json` (created on first alert config save):

```
{
  "enabled": true,
  "email": "user@gmail.com",
  "thresholds": {
    "zerodha": -5.0,
    "groww": -3.0
  },
  "baselines": {
    "zerodha": 125000.00,
    "groww": 48000.00
  },
  "last_triggered": {
    "zerodha": 1710234567.89
  }
}
```

Data Normalization

Both broker SDKs return data in different formats. The broker implementations normalize everything to a common schema before returning to the routers:

Holding (normalized):

Field	Type	Source (Zerodha)	Source (Groww)
`tradingsymbol`	`string`	`tradingsymbol`	`trading_symbol`
`exchange`	`string`	`exchange`	`"NSE"` (default)
`quantity`	`int`	`quantity`	`quantity`
`t1_quantity`	`int`	`t1_quantity`	`t1_quantity`
`total_quantity`	`int`	Computed	Computed

Field	Type	Source (Zerodha)	Source (Groww)
`average_price`	`float`	`average_price`	`average_price`
`last_price`	`float`	`last_price`	Via `get_ltp()`
`pnl`	`float`	`pnl`	Computed
`day_change`	`float`	`day_change`	`0` (limited API)
`broker`	`string`	`"zerodha"`	`"groww"`

Real-Time Data Flow

```
sequenceDiagram
    participant Frontend
    participant Backend
    participant Zerodha as Zerodha API
    participant Groww as Groww API

    loop Every 30 seconds (when market open)
        Frontend->>Backend: GET /api/portfolio/holdings
        Backend->>Zerodha: kite.holdings()
        Zerodha->>Backend: Raw holdings
        Backend->>Groww: groww.get_holdings_for_user()
        Groww->>Backend: Raw holdings
        Backend->>Backend: Normalize + merge + compute summary
        Backend->>Frontend: { holdings, summary }
    end
```

---

9. Market Hours and AMO Logic

NSE Market Timing

The market status engine (`backend/app/routers/market.py`) uses IST (Asia/Kolkata) timezone and defines these sessions:

Session	Time (IST)	Status	AMO Allowed
Before pre-open	00:00 -- 09:00	`closed`	Yes
Pre-open auction	09:00 -- 09:15	`pre_open`	No
Continuous trading	09:15 -- 15:30	`open`	No
Post-close	15:30 -- 15:40	`post_close`	No
After hours	15:40 -- 24:00	`closed`	Yes
Weekends	All day	`closed`	Yes
NSE holidays	All day	`closed`	Yes

Holiday Calendar

NSE holidays for 2026 are hardcoded in `NSE_HOLIDAYS_2026` :

```
NSE_HOLIDAYS_2026 = {
    "2026-01-26", "2026-03-10", "2026-03-17", "2026-03-30",
    "2026-04-01", "2026-04-03", "2026-04-14", "2026-05-01",
    "2026-05-25", "2026-07-07", "2026-08-15", "2026-08-26",
    "2026-10-02", "2026-10-20", "2026-10-22", "2026-11-04",
    "2026-11-09", "2026-12-25",
}
```

This must be updated annually with NSE's published holiday calendar.

AMO (After Market Order) Logic

AMO orders are allowed when the market status is `closed` and the current time is outside the pre-open to post-close window (i.e., before 09:00 or after 15:40). The frontend reads `allows_amo` from the market status response and adjusts the sell buttons accordingly:

- During market hours: regular sell orders
- Outside market hours (AMO eligible): sell buttons switch to AMO mode
- During pre-open/post-close: selling is disabled (neither regular nor AMO)

Frontend Polling

The `usePolling` hook refreshes holdings every 30 seconds, but only when:

- At least one broker is connected (`anyConnected === true`)
- The market is open (`marketStatus.is_open === true`)

When the market closes, polling stops automatically to avoid unnecessary API calls.

---

10. Alert and Notification System

Architecture

```
graph TD
    Startup["FastAPI Lifespan Start"] -->|"start_alert_loop()"| Thread["Daemon Thread<br/>(alert-monitor)"]
    Thread -->|"Every 60 seconds"| Check["check_alerts()"]
    Check --> LoadCfg["Load alerts_config.json"]
    Check --> TimeCheck{"Market hours?<br/>Mon-Fri 09:15-15:30"}
    TimeCheck -->|No| Skip["Skip"]
    TimeCheck -->|Yes| BrokerLoop["For each connected broker"]
    BrokerLoop --> ThresholdCheck{"Threshold configured?<br/>Baseline exists?<br/>Cooldown expired?"}
    ThresholdCheck -->|No| NextBroker["Next broker"]
    ThresholdCheck -->|Yes| FetchHoldings["Fetch holdings from broker"]
    FetchHoldings --> ComputeDrop["Compute portfolio drop %"]
    ComputeDrop --> DropCheck{"Drop exceeds threshold?"}
    DropCheck -->|No| NextBroker
    DropCheck -->|Yes| SendEmail["Send alert email via Gmail SMTP"]
    SendEmail --> UpdateCooldown["Update last_triggered timestamp"]
    UpdateCooldown --> SaveConfig["Save config to disk"]
```

Per-Broker Threshold Checking

Alerts operate at the **per-broker level**, not across the combined portfolio. Each broker has:

- An independent drop threshold (e.g., Zerodha: -5%, Groww: -3%)
- A baseline value (snapshot of portfolio value when alerts were enabled)
- An independent cooldown timer (1 hour between alerts for the same broker)

Cooldown Mechanism

After sending an alert for a broker, the system records `time.time()` in `last_triggered[broker_name]`. Subsequent checks skip that broker until `COOLDOWN_SECONDS` (3600 = 1 hour) have elapsed.

Email Delivery

Emails are sent via Gmail's SMTP server using SSL on port 465:

```
with smtplib.SMTP_SSL("smtp.gmail.com", 465) as server:
    server.login(smtp_email, smtp_app_password)
    server.send_message(msg)
```

Email content includes:

- Broker name and drop percentage
- Current portfolio value vs. baseline
- Drop amount in INR
- Top 3 losers by daily P&L
- Top 3 gainers by daily P&L

The `SMTP_EMAIL` must be a Gmail address, and `SMTP_APP_PASSWORD` must be a Gmail App Password (not the account password), generated from Google Account security settings.

---

11. Frontend Architecture

Component Hierarchy

```
graph TD
    App["App"] --> Header
    App --> Welcome["Welcome Screen<br/>(when no brokers connected)"]
    App --> Dashboard["Dashboard<br/>(when any broker connected)"]
    Dashboard --> BrokerTabs
    Dashboard --> FundsBar
    Dashboard --> PortfolioSummaryBar
    Dashboard --> HoldingsTable
    Dashboard --> SellButtons
    Dashboard --> OrderHistory
    Dashboard --> ConfirmModal
    Dashboard --> OrderResults
    Dashboard --> AlertConfig
    HoldingsTable --> BrokerBadge
    Header --> MarketBadge["Market Status Badge"]
```

Custom Hooks

Hook	API Call	Polling	Purpose
`useAuth`	`GET /api/auth/status`	On mount + after login/logout	Broker connection status
`usePortfolio`	`GET /api/portfolio/holdings`	Via `usePolling`	Holdings and summary data
`usePanicSell`	`POST /api/orders/panic-sell`	None	Execute sell orders
`useMarketStatus`	`GET /api/market/status`	Every 30s	Market open/closed status
`usePolling`	Calls `fetchHoldings`	Every 30s (when market open + connected)	Auto-refresh mechanism

State Management

The application uses **React local state and hooks only** -- no Redux, Zustand, or other external state libraries.

State	Location	Scope
Broker statuses	`useAuth` → `App`	Global
Holdings + summary	`usePortfolio` → `App`	Global
Selected holdings	`useState` in `App`	Dashboard
Broker filter tab	`useState` in `App`	Dashboard
Sell results	`usePanicSell` → `App`	Dashboard
Market status	`useMarketStatus` → `App`	Global
Modal state	`useState` in `App`	Dashboard

Filtered views (per-broker holdings, per-broker summary) are computed via `useMemo` based on the active broker filter tab.

API Communication

All API calls use the browser's native `fetch` API with relative `/api/` paths:

```
const res = await fetch("/api/portfolio/holdings");
const data: HoldingsResponse = await res.json();
```

- **Development**: Vite dev server proxies `/api/*` to `http://localhost:8000`
- **Production**: Vercel rewrites `/api/*` to the Render backend at the edge

No HTTP client library (Axios, etc.) is used.

UI Design System

Element	Implementation
Background	`surface-0` (#06060a) with SVG fractal noise overlay
Cards	Glassmorphism: `rgba(12,12,20,0.6)` + `backdrop-blur(20px)` + subtle borders
Fonts	Outfit (headings/body), JetBrains Mono (numbers/data)
Colors	Zerodha: `#387ed1`, Groww: `#00d09c`, Danger: `#ff3b5c`, Profit: `#00d68f`
Animations	`fade-in-up` (entrance), `scale-in` (modals), `glow-pulse` (panic button), `float` (decorative)
Stagger	`.stagger-1` through `.stagger-6` for sequential entrance animations

---

12. Security Considerations

Transport Security

- **HTTPS**: Enforced by both Vercel and Render at the edge. All traffic between the browser and both services is encrypted via TLS.
- **SMTP**: Uses SSL (port 465) for email delivery to Gmail.

CORS Policy

The backend restricts cross-origin requests to the configured `FRONTEND_URL`. In production, only the Vercel domain is allowed. Note that Vercel's rewrite proxy means most `/api` requests appear as same-origin to the browser, so CORS headers are primarily relevant for direct backend access.

Secrets Management

Secret	Storage	Access
Broker API keys/secrets	Render env vars	Server-side only
TOTP secrets	Render env vars	Server-side only
SMTP credentials	Render env vars	Server-side only
Broker access tokens	In-memory (Python)	Never sent to frontend

- `.env` is excluded from git (`.gitignore`) and Docker builds (`.dockerignore`)
- No secrets are embedded in frontend code
- Broker access tokens are never exposed to the client -- the frontend only receives a `connected: true/false` status

Session Security

- **No JWT, no cookies**: Sessions are server-side only, stored as Python object attributes. There is no session token sent to the client.
- **No session hijacking surface**: Since no session identifier is transmitted, there is nothing for an attacker to steal from the client side.
- **Trade-off**: The backend must be a single process (no horizontal scaling), and sessions are lost on restart.

Input Validation

- `variety` parameter is validated against `("regular", "amo")`
- Broker names are validated via `BrokerManager.get_broker()` which raises `ValueError` for unknown brokers
- `SellRequest` uses Pydantic model validation
- `AlertConfigRequest` uses Pydantic model validation

Rate Limiting

There is no rate limiting middleware. This is acceptable for a single-user tool but would need to be addressed for multi-user deployment.

SEBI Compliance

PanicSell is designed as a personal trading tool that complies with SEBI 2025 regulations:

- Uses only official broker APIs (Kite Connect, Groww Trade API)
- Does not engage in algorithmic trading or automated order placement without user initiation
- Every sell action requires explicit user confirmation via a typed challenge in the confirmation modal
- No third-party access to broker credentials

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13. Limitations and Future Considerations

Current Limitations

Limitation	Impact	Mitigation
Single-user architecture	Only one user per backend instance	Acceptable for personal use
No database	Alert config is the only persistent data; sessions lost on restart	Users re-authenticate daily anyway (token expiry)
Render free tier cold starts	First request after 15 min inactivity takes ~30 seconds	Acceptable for personal use; upgrade to paid tier ($7/mo) eliminates this
Hardcoded holiday calendar	Must be updated annually	Update `NSE_HOLIDAYS_2026` in `market.py` before 2027
No buy orders	Sell-only tool	By design -- scope limited to liquidation
Groww day change data	Limited to 0 (API doesn't expose intraday change)	Would require additional quote API integration
No LLM integration	No AI-powered features	N/A for current scope

Future Improvements

Improvement	Benefit
Redis for sessions	Survive process restarts, enable horizontal scaling
PostgreSQL for audit logs	Persistent order history, portfolio snapshots over time

Improvement	Benefit
Multi-user auth (JWT + database)	Support multiple users on a single deployment
Additional brokers (Angel One, Upstox)	Broader market coverage via the existing abstraction layer
WebSocket price streaming	Real-time price updates instead of 30-second polling
Dynamic holiday calendar	Fetch NSE holidays from an API instead of hardcoding
Portfolio analytics	Historical P&L charts, sector allocation, risk metrics
Mobile-responsive UI	Better experience on phone screens during market panics

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This document reflects the PanicSell architecture as deployed in March 2026. For the latest code, see the GitHub repository.